

GIRONA COSTA BRAVA AP, Spain

WMO: 081840

Lat: 41.912N

Lon: 2.763E

Elev: 129

StdP: 99.78

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.4	-1.1	-10.2	1.6	5.9	-7.0	2.1	4.6	10.4	8.1	8.6	11.0	0.6	10	0.424

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	34.0	21.3	32.4	21.1	31.1	20.8	23.4	29.9	22.7	29.0	22.0	28.1	5.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.6	16.6	25.7	20.8	15.7	25.4	20.0	14.9	25.0	70.2	30.0	67.4	28.9	64.8	28.1	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.9	7.7	6.5	-5.4	36.7	1.6	1.8	-6.6	38.0	-7.5	39.1	-8.4	40.1	-9.5	41.4		
			-6.2	24.9	1.7	0.9	-7.4	25.6	-8.4	26.1	-9.4	26.7	-10.6	27.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.4	7.9	8.4	10.9	13.3	17.0	21.4	24.1	24.1	20.4	17.0	11.5	8.5
	DBStd	6.47	2.78	2.85	2.75	2.59	2.77	2.97	2.38	2.34	2.59	2.95	3.28	2.86
	HDD10.0	244	77	57	22	4	0	0	0	0	0	1	22	62
	HDD18.3	1628	324	277	229	153	59	7	1	1	10	59	204	303
	CDD10.0	2226	11	13	51	101	216	343	436	438	313	219	68	17
	CDD18.3	567	0	0	0	1	17	100	178	179	73	19	0	0
	CDH23.3	5489	0	0	7	26	209	1033	1808	1714	559	132	1	0
	CDH26.7	2128	0	0	0	1	38	391	791	744	143	21	0	0
Wind	WSAvg	2.0	1.6	1.8	2.1	2.4	2.3	2.3	2.4	2.3	2.0	1.8	1.7	1.5
Precipitation	PrecAvg	690	47	44	55	62	62	42	31	43	83	102	63	56
	PrecMax	1035	166	122	150	181	159	243	108	86	166	301	280	178
	PrecMin	415	0	1	2	2	6	0	1	4	15	13	0	1
	PrecStd	173	44	37	38	44	41	47	28	24	41	70	56	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.3	20.7	24.0	25.9	30.1	35.2	35.9	36.7	31.6	29.1	22.7	19.2
		MCWB	12.2	12.3	14.3	15.8	18.4	20.9	21.5	21.7	20.8	18.8	15.7	12.8
	2%	DB	17.0	18.1	21.3	23.8	27.4	32.6	33.9	34.0	29.9	26.4	20.7	17.1
		MCWB	11.5	11.6	13.3	15.2	17.6	20.5	21.4	21.5	20.5	19.0	15.0	12.2
	5%	DB	15.1	16.4	19.3	22.0	25.6	30.8	32.4	32.2	28.2	24.7	19.0	15.7
		MCWB	10.3	10.8	12.5	14.4	17.0	19.8	21.3	21.3	19.9	18.3	14.2	11.5
	10%	DB	13.4	14.8	17.6	20.1	23.9	28.9	31.0	30.8	26.8	22.9	17.4	14.1
		MCWB	9.6	10.0	11.7	13.5	16.4	19.3	20.8	21.2	19.4	17.5	13.4	10.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.4	13.6	15.6	17.2	20.0	23.1	24.4	24.7	23.0	21.4	17.6	14.2
		MCDB	17.5	18.4	21.6	23.8	26.8	30.9	31.3	31.1	28.1	25.3	20.4	17.0
	2%	WB	12.2	12.3	14.2	15.9	18.7	21.7	23.3	23.6	22.1	20.2	16.5	13.1
		MCDB	15.4	16.7	19.6	22.5	25.4	28.9	30.2	29.9	27.2	24.1	19.0	15.8
	5%	WB	11.3	11.4	13.2	14.9	17.8	20.8	22.5	22.9	21.2	19.3	15.3	12.2
		MCDB	14.4	15.4	17.8	20.7	24.0	28.3	29.3	28.9	26.2	22.9	17.7	15.0
	10%	WB	10.2	10.5	12.4	14.0	16.9	20.0	21.7	22.2	20.3	18.3	14.1	11.2
		MCDB	12.9	14.0	16.5	19.0	22.6	27.3	28.5	28.0	25.1	21.7	16.6	13.9
Mean Daily Temperature Range	5% DB	MDBR	11.7	12.4	12.8	12.2	12.3	12.9	12.9	12.7	12.0	11.1	11.4	11.4
		MCDBR	14.1	15.0	15.8	15.3	15.0	15.5	14.8	15.0	14.0	13.4	12.9	12.8
		MCWBR	9.1	9.2	9.0	8.1	7.0	5.8	5.5	5.7	6.3	7.0	8.0	8.6
	5% WB	MCDBR	11.9	13.6	13.9	14.0	13.4	14.0	13.7	13.2	12.5	11.4	10.7	11.4
	MCWBR	8.2	8.8	8.4	8.0	7.0	6.1	5.8	6.1	6.4	6.6	7.2	8.2	
Clear-Sky Solar Irradiance	taub	0.314	0.332	0.377	0.402	0.421	0.431	0.437	0.432	0.410	0.383	0.342	0.314	
	taud	2.522	2.471	2.357	2.296	2.290	2.294	2.296	2.316	2.363	2.430	2.495	2.525	
	Ebn at Noon	839	875	867	868	857	846	836	830	822	802	794	805	
	Edh at Noon	72	88	110	126	130	130	128	122	109	91	74	66	
All-Sky Solar Radiation	RadAvg	1.97	2.88	4.13	5.16	6.03	6.86	6.79	5.97	4.56	3.18	2.17	1.74	
	RadStd	0.17	0.28	0.32	0.40	0.42	0.32	0.36	0.33	0.23	0.28	0.20	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+79	+52

Nomenclature: See separate page